

CYCLISTIC

CASE STUDY: Analyzing User Behavior

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INTRODUCTION

Cyclistic is a successful bike-share company based in Chicago with a fleet of over 5,800 bicycles and 600 docking stations. Until now, the company's pricing strategy relied on appealing to a broad consumer base through single-ride passes, full-day passes, and annual memberships.

However, Cyclistic's finance analysts have concluded that annual members are significantly more profitable than casual riders. The Director of Marketing believes the company's future success depends on maximizing the number of annual memberships. As a data analyst on the marketing team, my goal is to analyze historical bike trip data to identify trends in user behavior and help guide future marketing initiatives.

BUSINESS TASK

Primary Question: How do annual members and casual riders use Cyclistic bikes differently?

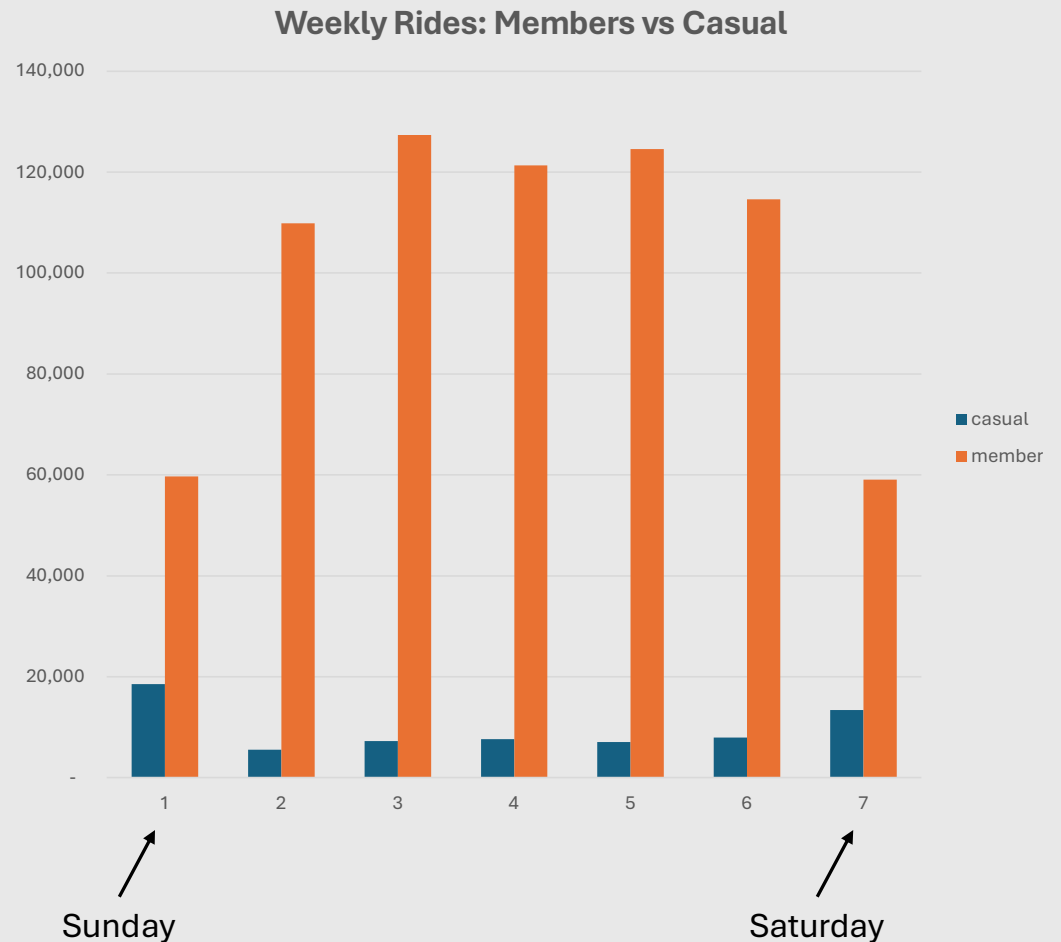
Objective: Analyze historical trip data to uncover actionable insights regarding user behavior and provide data-driven recommendations to help the marketing team design a targeted strategy to convert casual riders into annual members.

DATA SOURCE & RELIABILITY

- The data used for the analysis is historical, public bike trip data.
- The specific data analyzed for this report covers historical trip records from Q1 2019 and Q1 2020.
- Available by Motivate International Inc.
- Link: <https://divvy-tripdata.s3.amazonaws.com/index.html>
- The dataset is credible, reliable and appropriate for answering the core business questions of this case study.
- Strict data-privacy protocols were followed, ensuring that all personally identifiable information (PII) regarding the riders has been excluded.

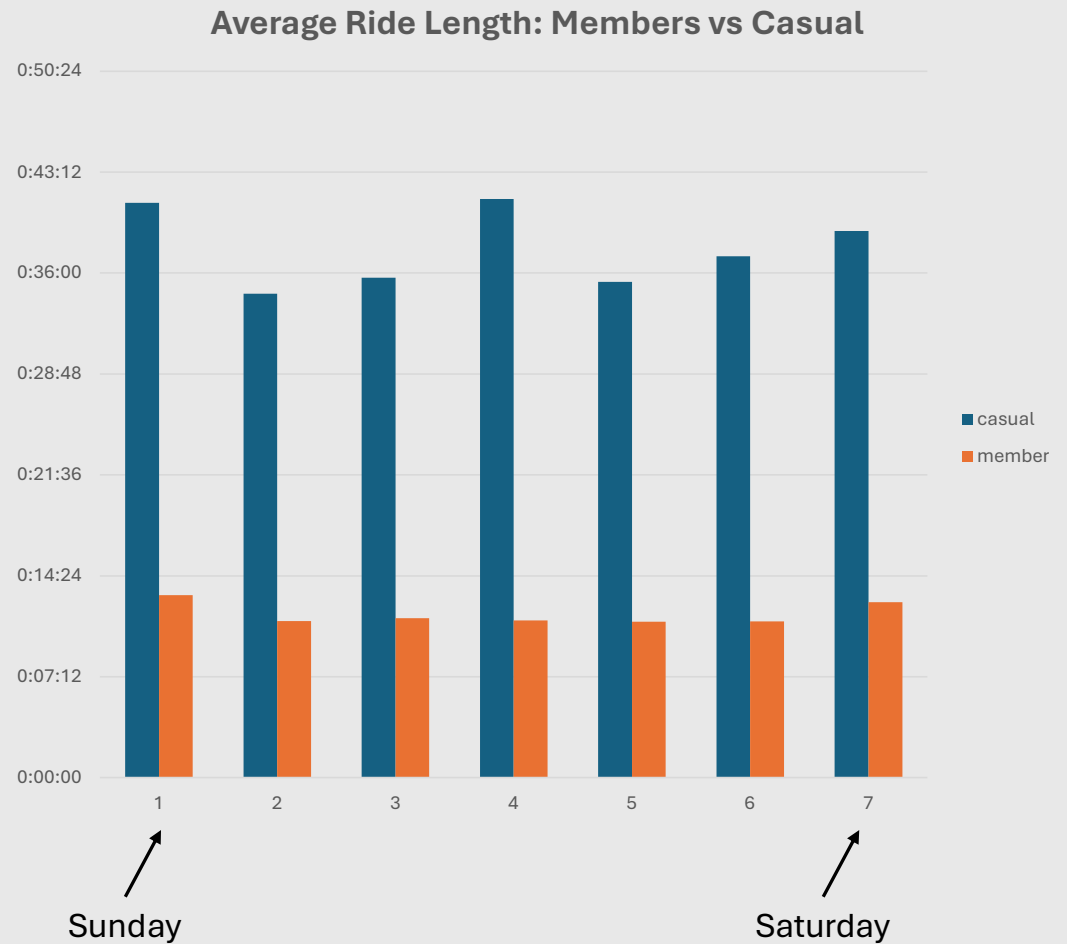
WEEKLY RIDE VOLUME

- **Key Takeaway:** Members commute during the week, while casual riders for leisure on weekends.
- **Member Consistency:** Annual members display high, consistent usage Monday through Friday, indicating they primarily use Cyclistic for daily commuting.
- **Casual Weekend Spikes:** Casual rider volume remains low during the workweek but nearly doubles on Saturdays and Sundays.



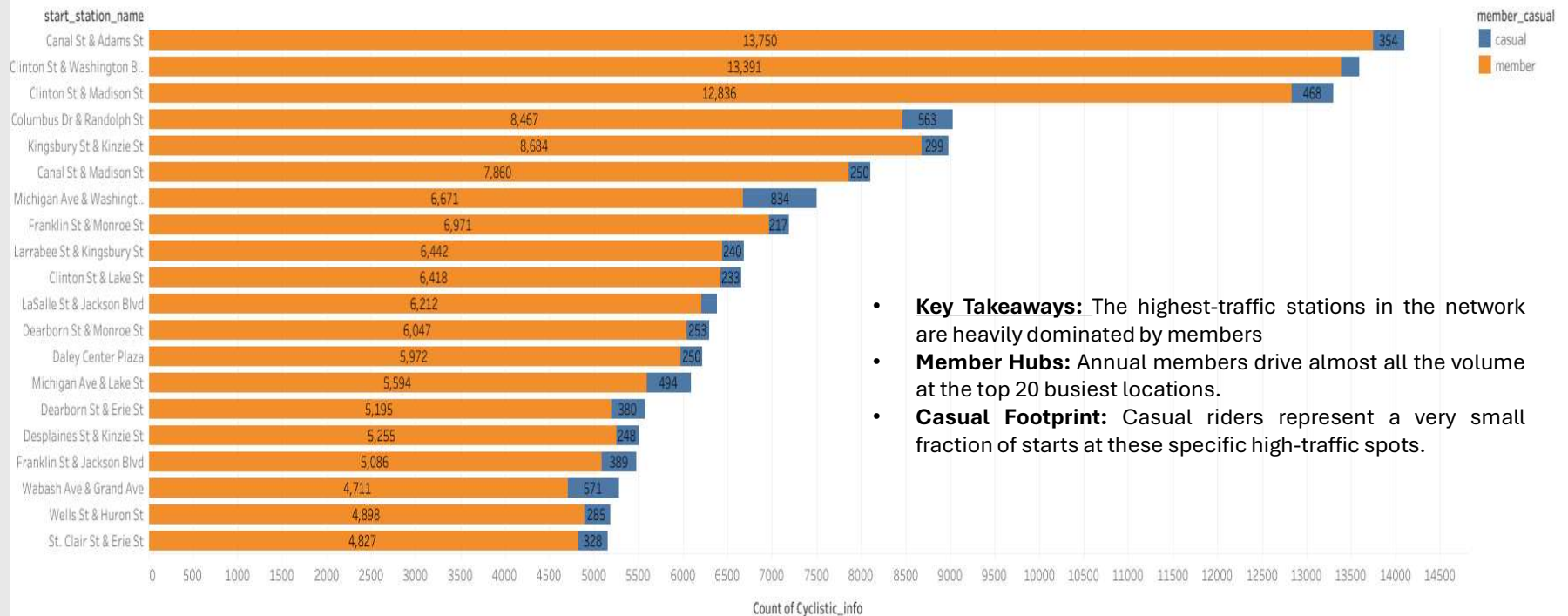
AVERAGE RIDE LENGTH

- **Key Takeaway:** Casual riders take trips nearly 3 times longer than annual members.
- **Casual Riders:** Average 35+ minutes per ride (recreational, leisure use).
- **Members: Average** ~12 minutes per ride (quick, point-to-point commutes).
- **Consistency:** This pattern remains consistent across all seven days of the week.



Top 20 Start Stations

Top 20 Start Stations



- **Key Takeaways:** The highest-traffic stations in the network are heavily dominated by members
- **Member Hubs:** Annual members drive almost all the volume at the top 20 busiest locations.
- **Casual Footprint:** Casual riders represent a very small fraction of starts at these specific high-traffic spots.

RECOMMENDATIONS

1. **Create a “Weekend-Only” Membership:** Offer a tailored annual plan specifically for riders who use bikes on Saturdays and Sundays.
2. **Advertise Cost Savings:** Launch a campaign showing casual riders how quickly an annual membership pays for itself on trips lasting longer than 30 minutes.
3. **Target Leisure Locations:** Focus physical and digital advertising near parks and recreational areas during peak weekend hours, rather than at commuter-heavy stations.

APPENDIX

Data Preparation & Cleaning Methodology

Data Aggregation & Standardization

- Merged the 2019 Q1 and 2020 Q1 datasets into a single workbook (merge_spreadsheet).
- Renamed 2019 Q1 columns (changing trip_id to ride_id and from_station_name to start_station_name) to align with the 2020 Q1 schema.
- Standardized user classifications in the 2019 Q1 data by updating “Subscriber” to “member” and “Customer” to “casual”.

Feature Engineering

- Generate a ride_length variable by subtracting the start time from the end time, formatted as HH:MM:SS.
- Created a day_of_week variable using the WEEKDAY formula to map start dates to numerical days (1=Sunday, 7=Saturday).

Data Cleaning & Quality Control

- Scanned for and confirmed zero duplicate entries in the ride_id column.
- Cleaned station names by removing asterisk (*) artifacts.
- Removed internal company testing linked to the “HQ QR” station.
- Eliminated outliers by excluding anomalous trips lasting less than 1 minute or longer than 24 hours.

THANK YOU